

Avi Garg

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EDUCATION

Thapar Institute of Engineering and Technology

B.E. in Electronics and Communication Engineering (CGPA: 8.53)

Relevant Coursework: Database Management Systems (DBMS), Data Structures and Algorithms (DSA), Design and Analysis of Algorithms (DAA), Operating Systems (OS)

Patiala, Punjab

Aug 2024 – May 2028

PROJECTS

Delivery Failure Predictor

Jupyter Notebook, React, Express, Vercel, Render

Hackathon Project

2026

- Developed an AI-powered delivery failure prediction system using a Machine Learning model trained on **Amazon delivery datasets** to identify orders at risk of unsuccessful delivery.
- Engineered a feature matrix of **61 features**, deriving domain-specific metrics like *traffic pressure*, *reliability risk*, and *distance-cost interactions* to evaluate complex logistics factors.
- Optimized a tuned **XGBoost Classifier** using **5-fold cross-validation**, achieving an overall validation accuracy of **96.18%**, an F1-score of **0.93**, and an AUC-ROC of **0.99**.
- Built an interactive React-based web application that connects via REST APIs to serve real-time delivery risk predictions to users.

Prodivity Technologies

React.js, Node.js, Express.js, MongoDB, Tailwind CSS

Freelance Agency Platform

2026

- Co-designed and developed a full-stack **freelance agency platform** to centralize developer portfolios and streamline **client acquisition** for a collective of peer developers.
- Engineered a secure **admin dashboard** enabling team members to dynamically **create, update, and manage projects and services** without manual code modifications.
- Designed and integrated **RESTful APIs** with **MongoDB** to handle high-performance content management, data persistence, and seamless frontend-backend synchronization.
- Developed a responsive **React.js user interface** that optimized asset loading and layout scalability, establishing a professional digital brand presence for the agency.

Malaria Detection System

Python, Resnet18, FastApi, React.js, Jupyter Notebook

AI for Engineers Course Project

2025

- Developed an **AI-powered malaria detection system** capable of classifying blood smear images as **Parasitized** or **Uninfected** using a **Deep Learning model**.
- Designed and trained the model on a curated malaria image dataset, performing **data preprocessing**, **image normalization**, and **model optimization** to improve classification performance.
- Built a responsive **React.js frontend** that allows users to upload microscopic cell images and receive **real-time diagnostic predictions**.
- Successfully **implemented, evaluated, and presented** the project as part of the **AI for Engineers** coursework, demonstrating the application of **Deep Learning** in **healthcare diagnostics and medical image classification**.

ACHIEVEMENTS

- Open Source & Version Control:** Managed and deployed **3+** full-stack production repositories on GitHub, utilizing advanced Git workflows, branching, and CI/CD tools.

SKILLS

Languages: C++, JavaScript, TypeScript, SQL, Python

Framework: React.js, Express.js, FastAPI, Node.js, Streamlit, Tailwind CSS

Database: MySQL, MongoDB(NoSQL)

System/Tools: Distributed Systems, System Design Fundamentals, Git, GitHub, VS Code